SOLVABILITY

The Rouché–Capelli Theorem

Existence of a solution, read off the ranks.

$A\mathbf{x} = \mathbf{b}$ is consistent

a solution exists

IMPLIES →

← CONVERSELY

$\operatorname{rank}(A) = \operatorname{rank}([A \mid \mathbf{b}])$

adding $\mathbf{b}$ as a column does not raise the rank

WHEN CONSISTENT

the solution set has dimension $n - \operatorname{rank}(A)$

Existence and dimension in a single rank comparison; A may be square, tall or wide. The proof: $\mathbf{b} \in \operatorname{Col}(A)$ exactly when the rank does not grow.